

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

#here we upload the image.....
image_path="/content/image.jpg"
image=cv2.imread(image_path)
print(image)
```

```
[[[9 7 6]
   [9 7 6]
   [9 7 6]
   ...
   [9 7 6]
   [9 7 6]
   [9 7 6]]]
```

```
[[[9 7 6]
   [9 7 6]
   [9 7 6]
   ...
   [9 7 6]
   [9 7 6]
   [9 7 6]]]
```

```
[[[9 7 6]
   [9 7 6]
   [9 7 6]
   ...
   [9 7 6]
   [9 7 6]
   [9 7 6]]]
```

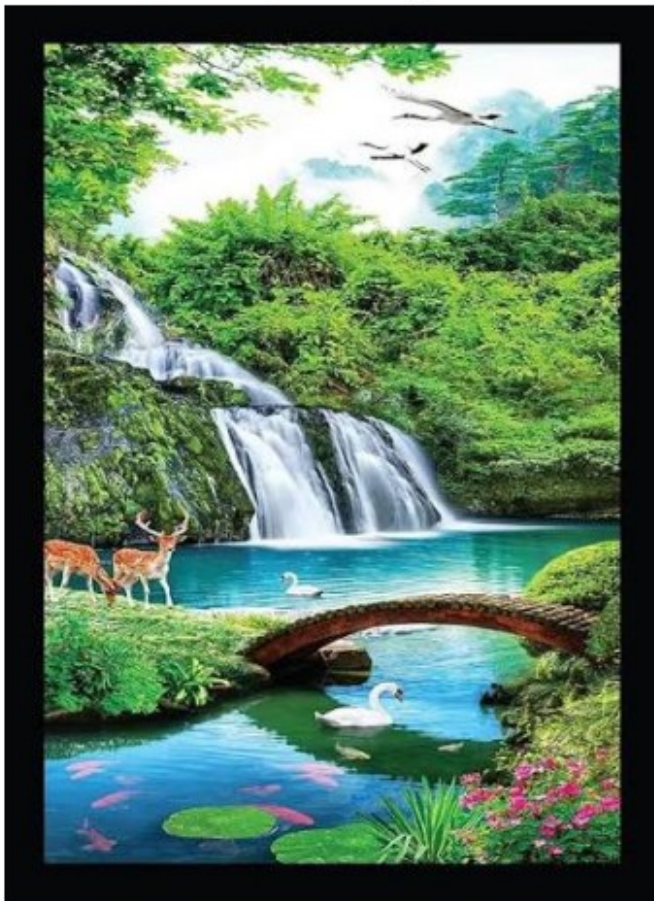
```
...
```

```
[[[9 7 6]
   [9 7 6]
   [9 7 6]
   ...
   [9 7 6]
   [9 7 6]
   [9 7 6]]]
```

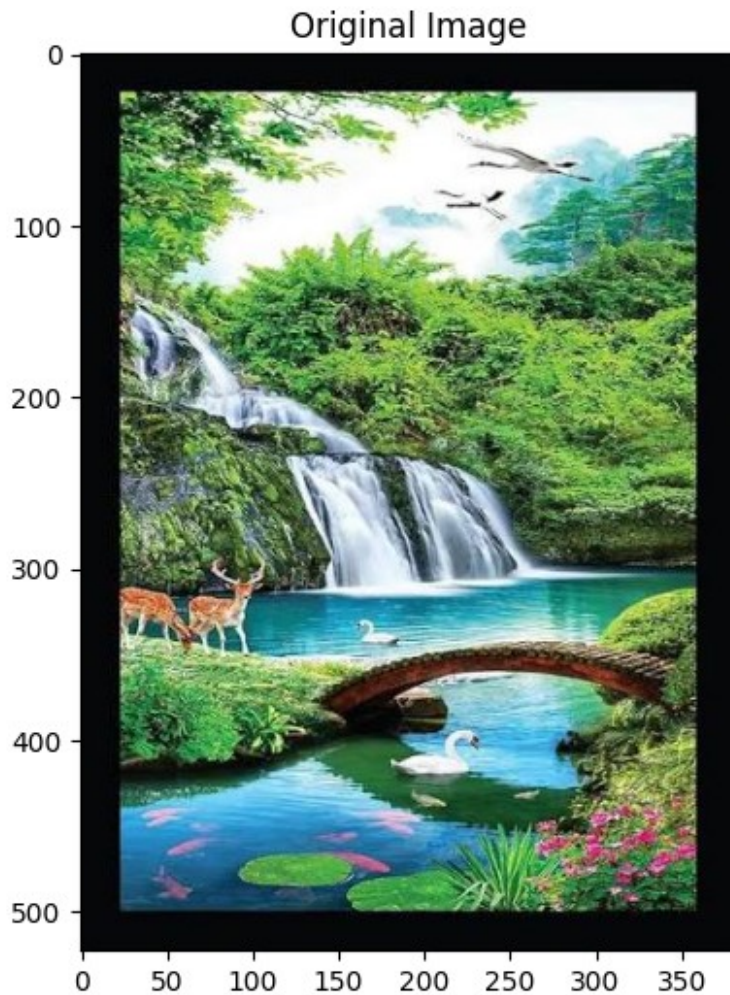
```
[[[9 7 6]
   [9 7 6]
   [9 7 6]
   ...
   [9 7 6]
   [9 7 6]]]
```

```
[9 7 6]]  
[[9 7 6]  
 [9 7 6]  
 [9 7 6]  
 ...  
 [9 7 6]  
 [9 7 6]  
 [9 7 6]]]
```

Original Image



```
image = cv2.imread(image_path)  
  
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)  
  
plt.figure(figsize=(8, 6))  
plt.imshow(image_rgb)  
plt.title("Original Image")  
  
plt.show()
```



```
print("Original Image Shape:", image.shape)
Original Image Shape: (523, 382, 3)

image_rgba = cv2.cvtColor(image, cv2.COLOR_BGR2RGBA)

image_gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("RGBA Image Shape:", image_rgba.shape)
print("Grayscale Image Shape:", image_gray.shape)

RGBA Image Shape: (523, 382, 4)
Grayscale Image Shape: (523, 382)

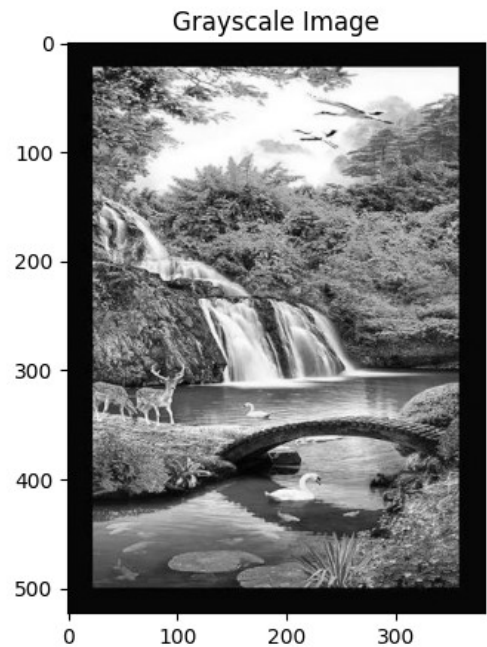
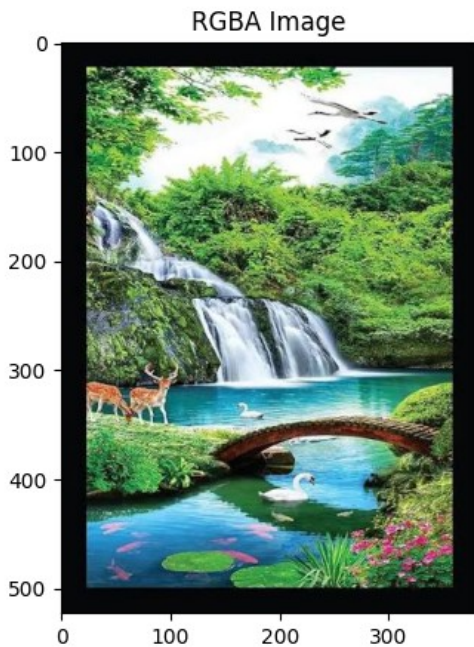
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(image_rgba)
```

```
plt.title("RGBA Image")

plt.subplot(1, 2, 2)
plt.imshow(image_gray, cmap="gray")
plt.title("Grayscale Image")

plt.show()
```



```
red_channel = image_rgb[:, :, 0]
green_channel = image_rgb[:, :, 1]
blue_channel = image_rgb[:, :, 2]

print("Red Channel Shape:", red_channel.shape)
print("Green Channel Shape:", green_channel.shape)
print("Blue Channel Shape:", blue_channel.shape)

Red Channel Shape: (523, 382)
Green Channel Shape: (523, 382)
Blue Channel Shape: (523, 382)

plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(red_channel, cmap="Reds")
plt.title("Red Channel")
plt.axis("off")

plt.subplot(1, 3, 2)
```

```
plt.imshow(green_channel, cmap="Greens")
plt.title("Green Channel")
plt.axis("off")

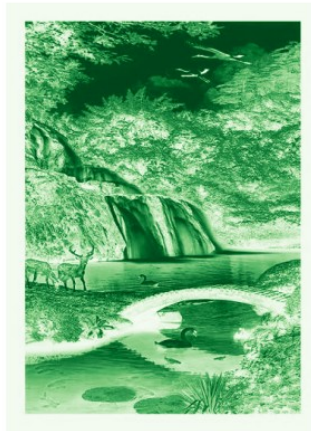
plt.subplot(1, 3, 3)
plt.imshow(blue_channel, cmap="Blues")
plt.title("Blue Channel")
plt.axis("off")

plt.show()
```

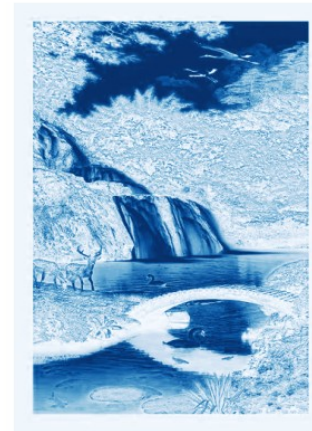
Red Channel



Green Channel



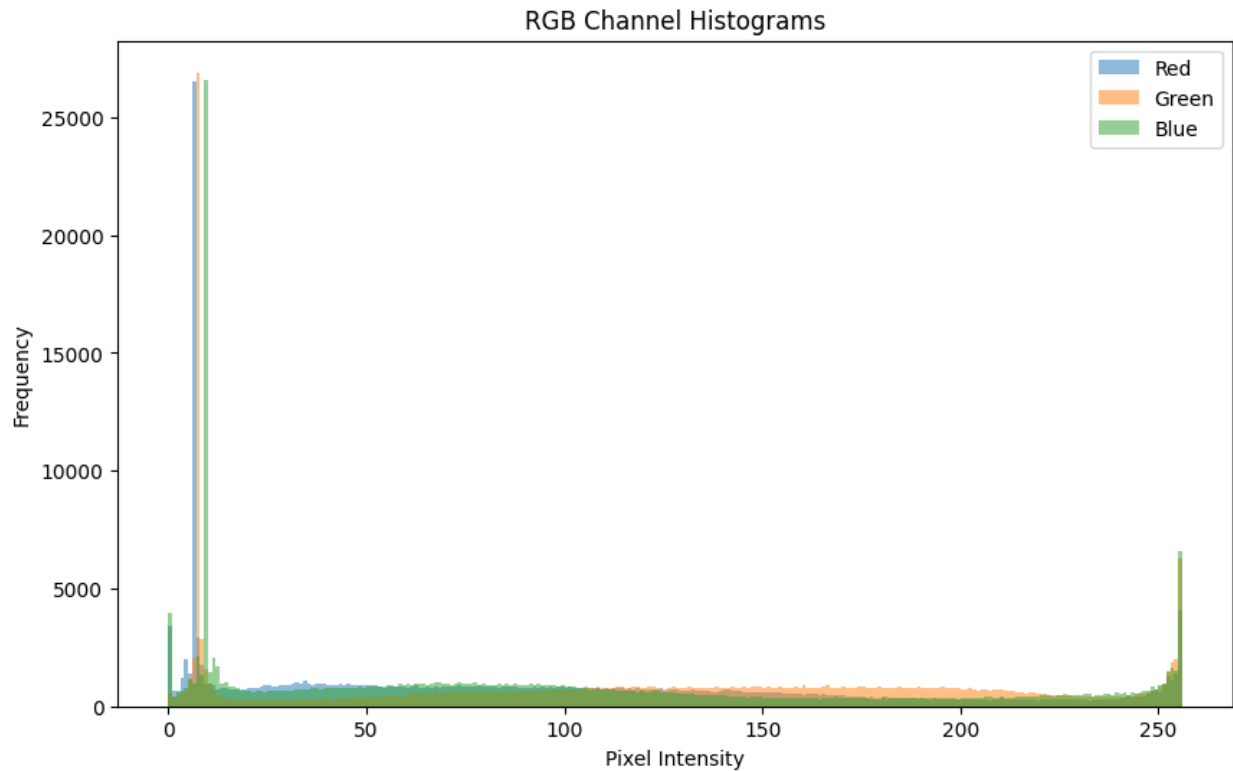
Blue Channel



```
plt.figure(figsize=(10, 6))

plt.hist(red_channel.ravel(), bins=256, range=[0, 256], alpha=0.5,
label="Red")
plt.hist(green_channel.ravel(), bins=256, range=[0, 256], alpha=0.5,
label="Green")
plt.hist(blue_channel.ravel(), bins=256, range=[0, 256], alpha=0.5,
label="Blue")

plt.title("RGB Channel Histograms")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```



```
image_ycrb = cv2.cvtColor(image_rgb, cv2.COLOR_RGB2YCrCb)

y, cr, cb = cv2.split(image_ycrb)

y_equalized = cv2.equalizeHist(y)

image_equalized_ycrb = cv2.merge([y_equalized, cr, cb])

enhanced_image = cv2.cvtColor(
    image_equalized_ycrb,
    cv2.COLOR_YCrCb2RGB
)

plt.figure(figsize=(14, 6))

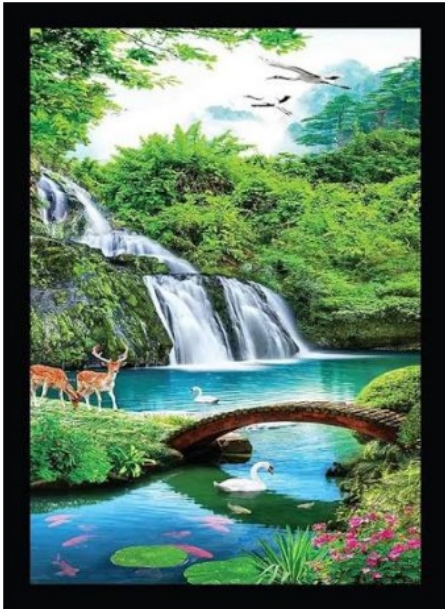
plt.subplot(1, 2, 1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(enhanced_image)
```

```
plt.title("Enhanced Image")
plt.axis("off")

plt.show()
```

Original Image



Enhanced Image



```
enhanced_red = enhanced_image[:, :, 0]
enhanced_green = enhanced_image[:, :, 1]
enhanced_blue = enhanced_image[:, :, 2]

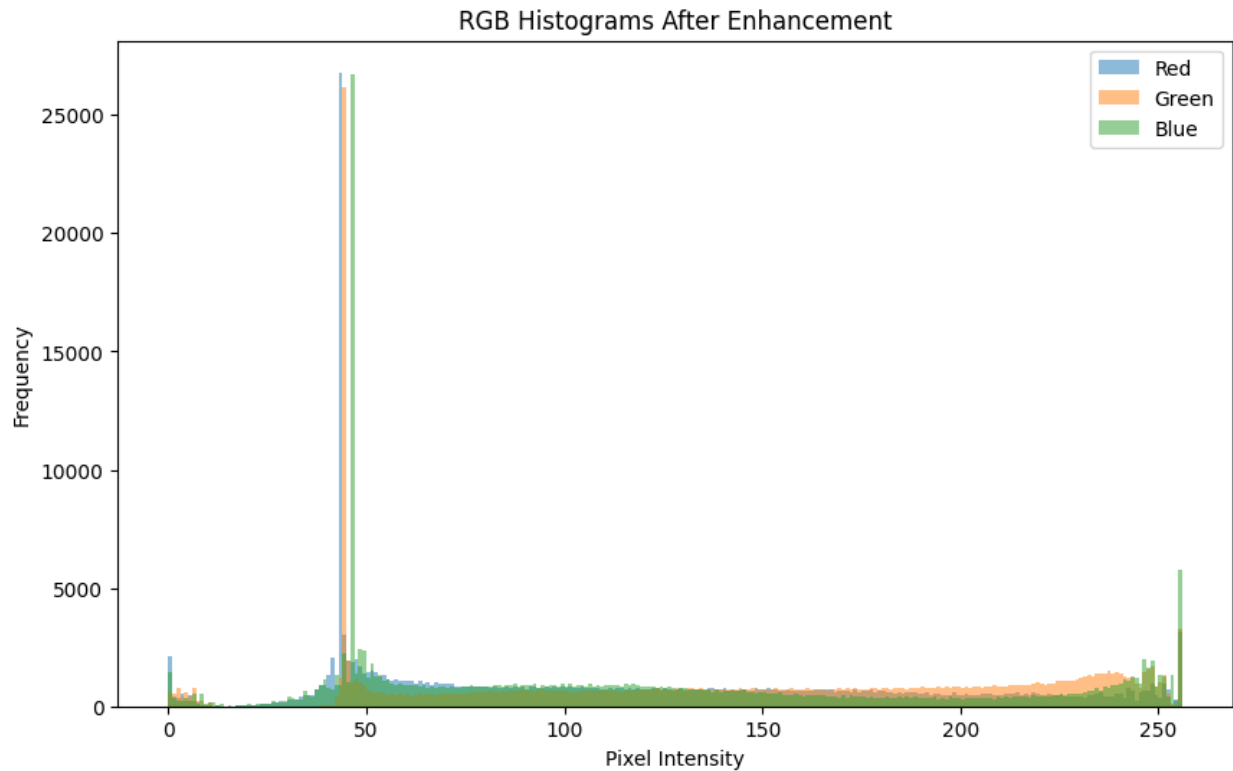
plt.figure(figsize=(10, 6))

plt.hist(enhanced_red.ravel(), bins=256, range=[0, 256],
         alpha=0.5, label="Red")

plt.hist(enhanced_green.ravel(), bins=256, range=[0, 256],
         alpha=0.5, label="Green")

plt.hist(enhanced_blue.ravel(), bins=256, range=[0, 256],
         alpha=0.5, label="Blue")

plt.title("RGB Histograms After Enhancement")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```



```
cv2.imwrite("original_image.jpg", image)
cv2.imwrite("rgba_image.png", cv2.cvtColor(image_rgba,
cv2.COLOR_RGBA2BGRA))
cv2.imwrite("grayscale_image.jpg", image_gray)
cv2.imwrite("red_channel.jpg", red_channel)
cv2.imwrite("green_channel.jpg", green_channel)
cv2.imwrite("blue_channel.jpg", blue_channel)
cv2.imwrite(
    "enhanced_image.jpg",
    cv2.cvtColor(enhanced_image, cv2.COLOR_RGB2BGR)
)
print("All images saved successfully!")
```

All images saved successfully!